

Austin Chung Khai Rui

Software Developer

✉ austinckr1229@outlook.com ☎ +6583183767 📍 Kuching, Malaysia

🔗 ackr-portfolio.site

🌐 <https://www.linkedin.com/in/austin-chung-khai-rui-940b92267/>

🐙 github.com/AustinCKR



👤 Profile

Passionate software developer with strong problem-solving skills and a keen interest in learning new technologies. Highly adaptable and eager to take on challenges, with hands-on experience in web, IoT and AI-based projects developed through teamwork and self-initiative.

🌐 Languages

Chinese

Native

English

Proficient

Malay

Basic

📁 Work Experience

03/2026 – Present
Singapore

Software Developer

United Microelectronics Corporation (UMC) Singapore

Deployed as a Software Developer at UMC Singapore via Hitec Sourcing.

🎓 Education

09/2022 – 09/2025

Bachelor of Computer Science

Swinburne University of Technology Sarawak ☑

Major: Software Development | **Minor:** Big Data Analytics

CGPA: 3.95 / 4.00

09/2021 – 08/2022

Foundation in Information Technology

Swinburne University of Technology Sarawak ☑

CGPA: 3.93 / 4.00

01/2016 – 12/2020

Sijil Pelajaran Malaysia (SPM)

SMK Sungai Maong

Grade: 1A+, 2A, 4A-, 2B+

📁 Projects

Digital Twin Simulation Platform for Fab Operations

- Designed and built a digital twin platform reconstructing fab-wide vehicle systems for data-driven simulation and analysis, loading historical data to replay operational conditions
- Used operationally by the team to diagnose vehicle congestion, enabling highlighting of congested vehicles and tracing their origin to identify root causes, directly supporting congestion resolution rather than serving as a passive visualization
- Built an interactive Dijkstra-based routing simulation enabling custom map edge modifications and side-by-side comparison of historical vs. simulated traffic, visualized through color-coded, thickness-weighted route overlays to identify and optimize high-traffic routes
- Integrated two structurally incompatible map systems (G2 & SRC) into a single unified visualization, decoding proprietary map file and vehicle transfer log formats to reconstruct the fab layout digitally

- Implemented 2D & 3D rendering using GDI+ to represent SRC's upper & lower track structure alongside G2's layout data in one combined view
- Extended the platform with additional operational data layers, including equipment status, FOUP input & output tracking, and grid lock visualization (automatic zone locking under congestion)

Customer Support System (Integration of WhatsApp, Facebook Messenger to a Web Application)

- Designed and developed analytics dashboards to track support ticket metrics and evaluate team performance across multiple messaging channels
- Built data processing and visualization features to support filtering, search, and reporting on large volumes of chat and ticket data
- Developed reusable data table components supporting advanced filtering, pagination, and multi-format data export
- Implemented role-based access control (RBAC) and user management modules to support operational scalability and security

Skills

Coding Languages & Frameworks

C#, Python, SQL, HTML, CSS, PHP, JavaScript,
DevExpress, Laravel, React JS, React Native, Tailwind CSS

Technical Skills

Git, Object Oriented Programming, Data Structures and Patterns,
AWS (EC2, RDS, S3), IoT systems (Arduino, Raspberry Pi, simple wiring and coding)

Organisations

04/2017 – 04/2020
Sarawak, Malaysia

St. John Ambulance Sarawak

First Aider

- Volunteer for charity run and sports competitions
- Person in charge of St. John Ambulance of SMK Sungai Maong for the Cheshire Home Charity Sale (Jul 2019), leading a team of 10

Awards

10/2025

Year 1 – Year 3 Six-Time Dean's List Recipient for All Semesters

Outstanding Performance in the Bachelor of Computer Science

NEUON AI Sdn Bhd

Top Scorer of Computing Technology Innovation Project & Intelligent Systems & IT Security

05/2025

Swinburne Faculty of Engineering and Computer Science Final Year Project 2025, Sem 1 'Best Group Award' Recipient

SOCOE Sdn Bhd

04/2025

Top Scorer of Software Engineering Project A

03/2024

Top Scorer of Data Structures and Patterns

08/2017

2017 Young Innovative Challenge Bronze Award Recipient